The exact source listed by ICMR is **"Standard Treatment Workflow (STW) for the Management of FEVER IN CHILDREN," ICD-10-R50**. ICMR currently lists it under **Paediatrics Standard Treatment Workflows**.

11. FEVER IN CHILDREN

Overview

Fever is a common clinical presentation in children and may occur due to a wide range of infectious and non-infectious conditions. The ICMR Standard Treatment Workflow emphasizes assessing the **duration of fever, associated/localizing symptoms, general appearance, vital signs and physical findings** to identify the likely cause and determine whether investigations, treatment, hospitalization or referral are required.

Definition of Fever

According to the ICMR STW:

- **Core/rectal temperature ≥38.0°C (100.4°F)** is considered fever.
- **Axillary temperature >37.5°C** is also considered fever.

What to Ask

When evaluating a child with fever, the ICMR workflow recommends asking about:

- **Duration of fever**
- Localizing symptoms suggesting:
    - Respiratory tract infection
    - Urinary tract infection
    - Gastrointestinal infection
    - Central nervous system infection
- **Rash**
- **Joint symptoms**
- Skin or soft-tissue swelling/redness
- Recent vaccination within 24 hours
- Drug or toxin exposure
- Similar illness among family members or in the neighbourhood.

Clinical Examination

Vital Signs

Assess:

- Temperature
- Heart rate
- Respiratory rate
- Blood pressure
- Capillary filling time (CFT)

General Appearance

Look for:

- Sick or toxic appearance
- Lethargy
- Irritability
- Inconsolability
- Dehydration

General Examination

Examine:

- Ear, nose and throat
- Skin and rash
- Lymph nodes
- Joints
- Genital area
- Bones
- Skin for pustules, pyoderma, impetigo or cellulitis.

Systemic Examination

The workflow includes examination of:

- Chest
- Abdomen

- Central nervous system
- Cardiovascular system.

Clinical Clues to Possible Diagnosis

The ICMR workflow provides specific symptom combinations that can help identify possible causes.

Fever + Respiratory Symptoms

- **Cough + runny nose →** Upper respiratory tract infection (URTI)
- **Membrane over tonsils/pharynx →** Diphtheria
- **Paroxysmal cough →** Pertussis-like illness
- **Barking cough →** Laryngotracheobronchitis / croup

Fever + Rash

- **Red maculopapular rash →** Measles, rubella or dengue
- **Fine generalized maculopapular rash + systemic dysfunction/shock →** Meningococcemia
- **Itchy erythematous macules progressing to clear vesicles →** Varicella

Fever + Other Findings

- **Parotid swelling →** Mumps
- **Arthritis →** Consider chikungunya, acute rheumatic fever or juvenile idiopathic arthritis
- **Strawberry-red tongue + skin peeling + lymphadenopathy + conjunctival injection →** Kawasaki disease.

Investigation of the Febrile Child

The ICMR STW emphasizes that investigations should be performed **when the result is expected to influence management**.

Fever <7 Days

For a child with **fever alone**:

- If fever is **<72 hours** and the child does not appear sick:

    - **No investigations are generally required.**
- If fever persists for **>72 hours**, investigations may include:
    - Total leukocyte count (TLC)
    - Differential leukocyte count (DLC)
    - Peripheral smear for leukocyte morphology
    - Malarial parasite testing
    - Platelet count

Additional investigations may be considered according to the clinical situation.

Fever >7 Days

When fever persists for more than 7 days, particularly when **localizing symptoms are present**, broader investigations may be considered based on clinical findings.

The ICMR workflow lists investigations such as:

- Serological testing for selected infections
- Bone marrow examination where indicated
- ANA profile
- HIV serology
- Echocardiography
- CT/PET imaging in selected situations
- Testing for infections such as **Brucella, CMV, herpes and Japanese encephalitis**, depending on the clinical presentation.

Important: These are **not tests that every child with fever should undergo**. The ICMR workflow uses them selectively according to fever duration, symptoms and clinical suspicion.

Management

The ICMR workflow organizes management into **five steps**.

STEP 1 — Initial Assessment and Stabilization

- Identify and manage urgent problems.
- Reduce fever using **paracetamol (15 mg/kg by any route)** and/or hydrotherapy.

- Manage life-threatening conditions immediately.
- Consider the first dose of an antibiotic when meningitis, severe pneumonia or severe malnutrition is suspected.
- Consider antimalarial treatment when malaria is suspected.

STEP 2 — Consider Hospitalization

Hospital observation/management should be considered for:

- **All neonates**
- Young infants with a toxic appearance
- Children with severe malnutrition
- Toxic appearance
- Inability to feed
- Lethargy
- Irritability
- Dehydration
- Fever/illness lasting **>14 days without a diagnosis**
- Any other situation where the treating physician considers hospitalization necessary.

STEP 3 — Consider Referral to a Tertiary-Care Centre

After appropriate stabilization and/or initial management, referral should be considered when there is:

- Need for intensive care
- Complex multisystem disease
- Confirmed complications of the primary illness.

STEP 4 — Empiric Management

Empiric treatment should be based on:

- Likely diagnosis

- Clinical severity/sickness status
- Availability of investigation facilities

Treatment should subsequently be **modified according to investigation results and local antimicrobial sensitivity patterns**.

The ICMR STW specifically states that **anti-tuberculosis treatment should not be started empirically**, except in suspected tuberculous meningitis.

STEP 5 — Consider Discharge When

Discharge may be considered when:

- The child has been **afebrile for >48 hours** or fever is showing defervescence.
- The child is feeding well.
- Symptoms other than fever have resolved or are resolving.
- The physician is satisfied that further care can safely continue on an ambulatory basis.
- The required duration of IV antibiotic therapy has been completed.

When Urgent Medical Attention Is Needed

Based on the ICMR workflow, particular concern is warranted when a febrile child has:

- Toxic or very sick appearance
- Lethargy or marked irritability
- Inability to feed
- Dehydration
- Signs suggesting meningitis or severe pneumonia
- Severe malnutrition
- Signs of systemic dysfunction or shock
- Complex multisystem involvement
- Prolonged fever without an established diagnosis.

Quick Fact Sheet

Feature	Fever in Children
ICMR source	Standard Treatment Workflow (STW)
Scope	Children
ICD-10	R50
Fever definition	Core ≥38.0°C or axillary >37.5°C
Initial assessment	Fever duration + symptoms + vital signs + clinical examination
Common clues	Respiratory symptoms, rash, joint symptoms, GI/CNS symptoms
Investigations	Based on duration, symptoms and whether results affect management
Initial treatment	Stabilization + fever control + condition-specific management
Hospitalization	Based on age, appearance, feeding, dehydration, severity and duration
Referral	Intensive-care need, complex multisystem disease or complications
Discharge	Afebrile/improving, feeding well, symptoms resolving and physician satisfied

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VISUALS YOU SHOULD ADD

For this particular topic, **the ICMR STW itself is already essentially a flowchart**, so this is different from #9 and #10.

1. ⭐ MOST IMPORTANT — Recreate the ICMR workflow

Use this structure:

CHILD WITH FEVER
↓
WHAT TO ASK?
↓
EXAMINATION
↓
CLUES TO SPECIFIC DIAGNOSIS

↓

DECIDE INVESTIGATIONS

↓

INITIAL STABILIZATION

↓

HOSPITALIZATION / REFERRAL IF REQUIRED

↓

EMPIRIC / CONDITION-SPECIFIC MANAGEMENT

↓

DISCHARGE WHEN CRITERIA ARE MET

This will make your PDF much more useful for the **AI health-awareness chatbot** project.

2. Fever + Symptoms visual

You can make a simple central diagram:

FEVER

→ Respiratory symptoms
→ Rash
→ Joint symptoms
→ GI symptoms
→ CNS symptoms
→ Skin/soft-tissue findings

Then show examples of possible diagnoses from the ICMR STW.

3. Investigation timeline

A very useful visual would be:

<72 hours + child not sick → Usually no investigation

>72 hours → Consider basic investigations

>7 days / localizing symptoms → Targeted investigations based on clinical findings

This is directly based on the ICMR workflow

Official reference

Indian Council of Medical Research (ICMR), Department of Health Research, Ministry of Health & Family Welfare, Government of India. *Standard Treatment Workflow (STW) for the Management of Fever in Children, ICD-10-R50*.

ICMR — Standard Treatment Workflows (official page)